

RESEARCH ARTICLE

Implanting Atomically Dispersed Fe and Mo Sites into 3D Scaffolds: A Scalable Route to Bifunctional Water Splitting Electrocatalysts

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Received: 24 March 2026 | **Revised:** 18 May 2026 | **Accepted:** 9 June 2026

Keywords: 3D porous scaffolds | ion implantation | single-atom catalysts | water splitting

ABSTRACT

Due to geometric constraints of planar substrates and poor electrical interface properties, overcoming the performance bottlenecks of single-atom catalysts (SACs) in industrial water splitting remains a challenge. Here, we extend ion implantation technology to hierarchical three-dimensional (3D) structures by constructing iron- and molybdenum-based SACs uniformly anchored on carbon nanotube/nickel foam (CNT/NF) scaffolds. This physical approach enables the precise and scalable incorporation of metal atoms with high dispersion and accessibility on a scale of up to 8 inches. Implantation-induced M–C coordination optimizes the electronic structure of active sites, lowering the reaction energy barrier by shifting the d-band center and enhancing d-p orbital hybridization. Concurrently, the 3D CNT/NF scaffold facilitates rapid charge transfer and efficient bubble release. An anion exchange membrane (AEM) electrolyzer with the two catalysts delivered a current density of 500 mA cm^{−2} at 1.66 V and 60°C and demonstrated robust durability over 100 h of continuous operation. This work establishes a scalable, effective platform for designing high-performance, durable non-precious SACs for practical water electrolysis.

1 | Introduction

The depletion of fossil fuels and their environmental impact have sparked widespread interest in hydrogen production via water electrolysis, which is viewed as a sustainable, zero-carbon energy solution [1]. Among various electrolysis technologies, alkaline water electrolysis is considered one of the most industrially promising hydrogen production solutions due to its technological maturity, long service lifetime, and relatively low scaling costs [2]. However, its practical application is still limited by insufficient energy conversion efficiency and long-term durability [3]. Although noble metal-based catalysts can significantly enhance electrolysis performance, their scarcity and high cost severely limit large-scale application [4–6]. Therefore, developing effi-

cient and durable electrocatalysts based on earth-abundant and non-precious metals is essential [7–9].

Modulating the electronic structure of catalytic active sites through interfacial electronic effects has emerged as an effective strategy to enhance electrocatalytic performance [10, 11]. In particular, anchoring non-precious metals as single atoms on suitable supports enables efficient tuning of the intrinsic electronic structure via strong metal-support interaction [12, 13]. Recent studies have demonstrated that self-supported non-precious metal electrodes can achieve high activity and long-term stability for water electrolysis by improving mass transport and active-site accessibility [14, 15]. Moreover, synergistic interactions between metal sites and supports have been shown to

optimize electronic structure and enhance catalytic performance, further demonstrating the importance of interfacial cooperation [16]. Such modulation operates through three primary mechanisms. First, charge transfer between the metal center and the support redistributes electron density at active sites, thereby facilitating water adsorption and dissociation [17–19]. Second, d-band center modulation adjusts the energy position of metal d orbitals relative to the Fermi level, optimizing key intermediate binding [20]. Third, orbital hybridization between metal d orbitals and supports or adsorbate p orbitals enhances electron coupling, strengthening metal-reactant interactions and lowering reaction energy barriers [21, 22]. These mechanisms have been validated in representative systems, including a Cu-M_xO_y catalyst, where electronic metal-support interactions induce charge transfer from Cu to the oxide supports, modulating the Cu d-band and enhancing reactant adsorption and activation [23], and Ce single-atom on ZnO, where strong d–p orbital hybridization promotes *COOH formation and accelerates CO generation [24].

Transition metals such as Fe, Co, Ni, and Mo feature tunable d-band electronic structures [25, 26], and their hybridization with the s/p orbitals of low-dimensional carbon supports (e.g., graphene, CNTs, N-doped graphene) allows precise regulation of intermediate adsorption [27, 28]. This combination of unsaturated coordination environment and unique electronic structure endows non-precious metal single-atom catalysts (SACs) with strong potential to replace noble metal-based catalysts [29, 30]. Indeed, nickel single atoms on three-dimensional porous graphene have demonstrated high activity and stability in the hydrogen evolution reaction (HER), largely due to electron transfer induced by sp-d orbital hybridization between Ni atoms and surrounding C atoms [31]. Despite these advances, practical electrode fabrication remains challenging. Conventional “synthesize-then-coat” strategies often cause active-site blockage or aggregation [32, 33], while in situ growth approaches are limited by harsh synthesis conditions, low yield, and poor process versatility [34–36]. Therefore, developing self-supported single-atom catalyst electrodes that combine high intrinsic activity with structural integrity at the macroscopic scale is crucial for advancing single-atom catalysts toward practical application.

Recently, ion implantation has emerged as a robust physical strategy to overcome the thermodynamic instability of SACs, as demonstrated by the precise stabilization of Co single atoms on oxide surfaces [37]. However, the previously reported methods rely on two-dimensional (2D) planar substrates, which may limit the overall active-site density and hinder gas-bubble release during high current density operation. Consequently, the macroscopic performance of such ion-implanted catalysts often fails to reach their full potential, underperforming compared to catalysts synthesized via conventional chemical routes.

In this study, we extended this methodology to hierarchical three-dimensional (3D) architectures by constructing iron- and molybdenum-based single-atom catalysts on a carbon nanotube/nickel foam (CNT/NF) composite substrate (denoted as Fe SACs/CNT/NF and Mo SACs/CNT/NF, respectively), which were selected as the optimal candidates for the oxygen evolution reaction (OER) and HER, respectively, after an initial screening of various transition metals (Fe, Co, Cu, Mo, as shown in Figure S1) under identical conditions. This strategy ensures

highly dispersed and robust atomic loading of Fe and Mo atoms within the macroscopic conductive scaffold, enabling the scalable fabrication of self-supported catalytic electrodes up to 8 inches, with high wafer-scale uniformity verified by reproducible HER performance across multiple 2 × 2 cm² samples, and is fully compatible with semiconductor industry-standard ion implantation processes for high-throughput, wafer-scale production [38–40]. Structural analysis and density functional theory (DFT) calculations confirm that implantation-induced M–C bonding and coordination reconfiguration optimize the electronic structure by tuning the d-band center and enhancing d-p orbital hybridization. These effects collectively optimize intermediate adsorption, yielding exceptional overpotentials of 14 mV for the HER and 260 mV for the OER at 10 mA cm^{−2}. Consequently, the 3D bifunctional electrodes achieve an industrial-level current density of 500 mA cm^{−2} at 1.66 V in an anion exchange membrane (AEM) electrolyzer, with over 100 h of durability. Techno-economic analysis further highlights its commercial viability, with a projected hydrogen cost of ~\$2.94 per kg H₂—41% lower than noble-metal benchmarks (approximately \$5.01 per kg H₂), as detailed in Tables S1 and S2. These results not only highlight the potential of this class of catalysts for cost-effective and efficient large-scale clean hydrogen production but also provide an industrially promising technical pathway for the structural design and controllable preparation of non-precious metal catalysts.

2 | Results and Discussion

2.1 | Synthesis and Characterization of Fe- and Mo-Based SACs

The Fe SACs/CNT/NF and Mo SACs/CNT/NF catalysts were prepared by first dip-coating carbon nanotubes (CNT) onto nickel foam (NF) to form a conductive three-dimensional scaffold, followed by precise ion implantation of the target metals under vacuum at room-temperature (Figure 1). Inductively coupled plasma optical emission spectroscopy (ICP-OES) measurements indicate Fe and Mo loadings of 0.82 and 0.88 wt.%, respectively. These loadings correspond to the optimal implantation dose (~1 × 10¹⁷ ions cm^{−2}) identified in preliminary studies (Figure S2), which balances the density of single-atom active sites with the structural integrity of the CNT support, thereby maximizing catalytic activity while minimizing potential degradation. All subsequent characterizations and electrochemical measurements were performed using these optimized catalysts. Scanning electron microscopy (SEM) images reveal that the NF surface becomes a roughened texture coated with a cross-linked CNT network (Figure 2a and Figure S3). The continuous CNT network not only ensures electrical interconnectivity throughout the scaffold (Figure 2b) but also provides abundant anchoring sites for metal atoms. Aberration-corrected high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images clearly display isolated bright dots on the CNT surfaces, confirming the atomic dispersion of Fe species (highlighted by red circles in Figure 2c). Scanning transmission electron microscopy (STEM) images and energy-dispersive X-ray spectroscopy (EDS) mapping further confirm the uniform distribution of Ni, Fe (or Mo), and C after ion implantation (Figure 2d,e). These results demonstrate the successful synthesis

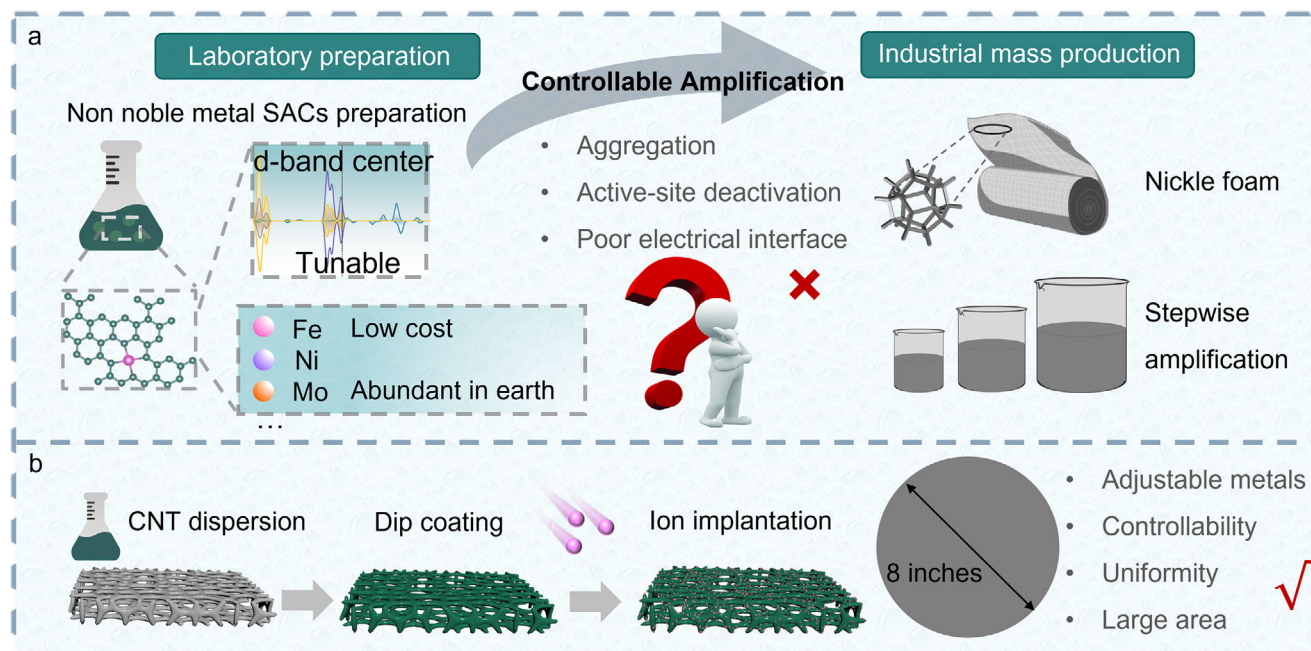


FIGURE 1 | Schematic illustration of a controllable synthesis and amplification strategy for transition metal SACs. (a) Solution-based methods for SACs synthesis face a critical challenge during scale-up. (b) Ion implantation for SACs fabrication: process and advantages.

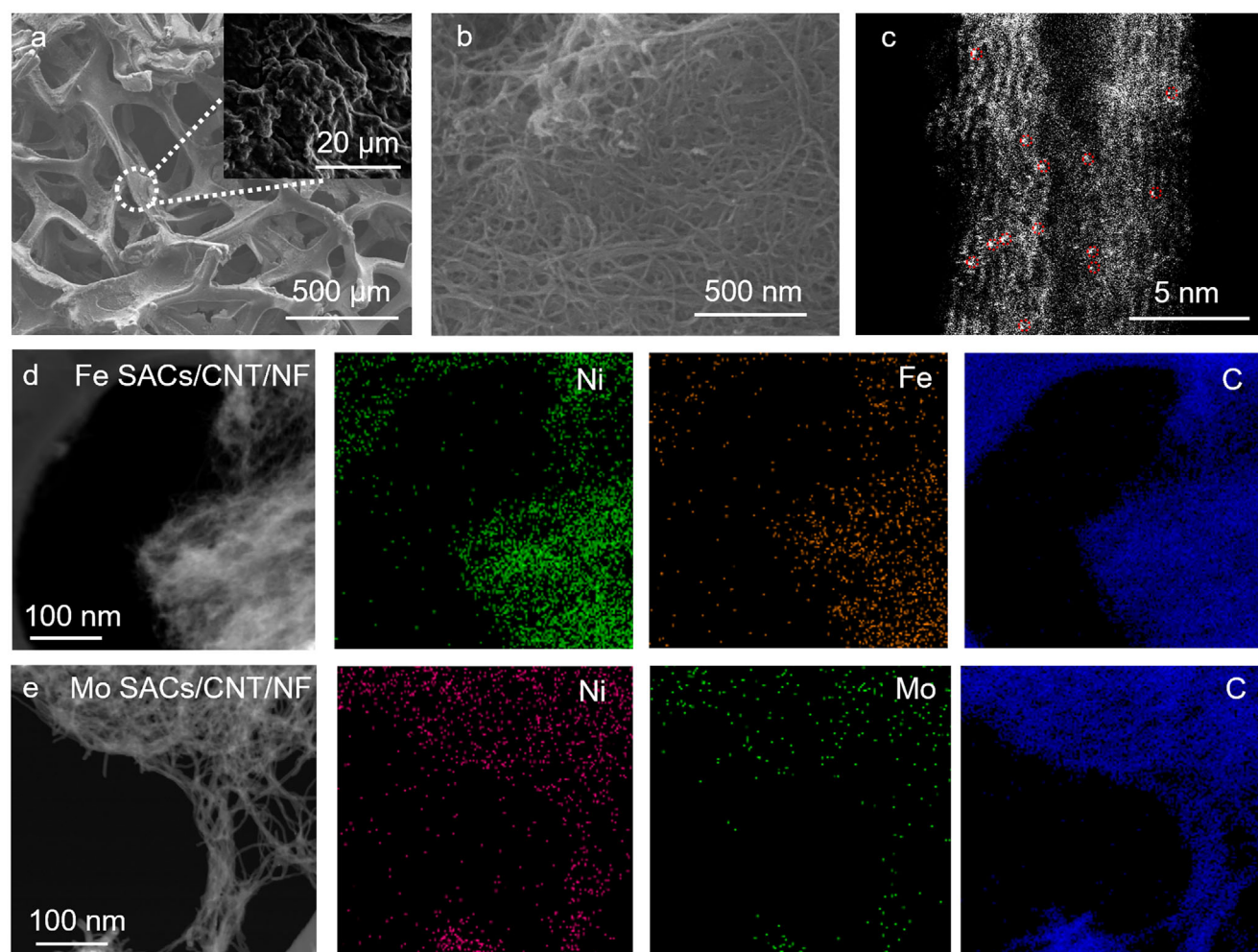


FIGURE 2 | Morphology of Fe- and Mo-based SACs. (a, b) SEM images of Fe SACs/CNT/NF. (c) Aberration-corrected HAADF-STEM image of Fe SACs/CNT/NF. STEM images and EDS mapping of corresponding elements of (d) Fe SACs/CNT/NF and (e) Mo SACs/CNT/NF.

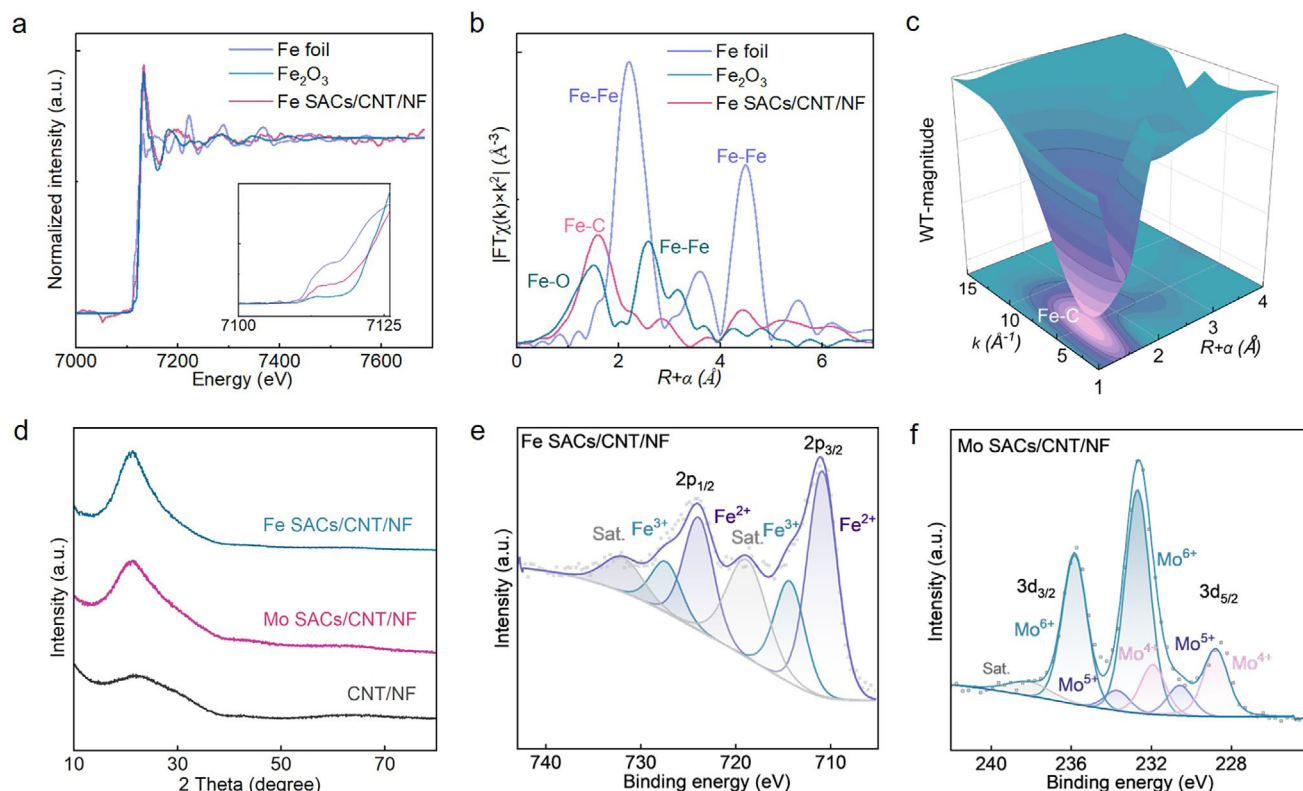


FIGURE 3 | Structural analysis of Fe- and Mo-based SACs. (a) XANES spectra of Fe K-edge. (b) FT-EXAFS spectra of Fe K-edge. (c) WT-EXAFS contour maps of the Fe K-edge. (d) XRD patterns of Fe SACs/CNT/NF, Mo SACs/CNT/NF, and CNT/NF. High-resolution XPS spectra of (e) Fe 2p in the Fe SACs/CNT/NF and (f) Mo 3d in the Mo SACs/CNT/NF.

of Fe SACs/CNT/NF and Mo SACs/CNT/NF catalysts, with target metal atoms effectively implanted into CNT-coated NF.

The chemical states and atomic coordination structures of the catalysts were investigated using X-ray absorption near-edge spectroscopy (XANES) and extended X-ray absorption fine structure (EXAFS) spectroscopy. As shown in Figure 3a, the near-edge absorption energy of Fe SACs/CNT/NF lies between those of the Fe foil and Fe₂O₃ references, indicating an oxidized state between Fe⁰ and Fe³⁺. The corresponding Fourier transform (FT) k²-weighted EXAFS spectrum from Figure 3b shows a primary peak at approximately 1.59 Å, assigned to the first-shell Fe–C coordination. Notably, the absence of the Fe–Fe peak indicates that Fe atoms are atomically dispersed within the CNT/NF matrix [41–43], consistent with the HAADF-STEM observations in Figure 2c. Due to the strong resolution in both k and R spaces, the wavelet transform (WT) of the Fe K-edge EXAFS oscillations was employed to further explore the atomic dispersion of Fe in the Fe SACs/CNT/NF, as shown in Figure 3c. Compared with the WT maps of Fe foil and Fe₂O₃ (Figure S4), no maximum intensity corresponding to Fe–Fe contributions was detected. EXAFS fitting results (Figure S5 and Table S3) show that each Fe atom is coordinated with five C atoms at a bond length of 2.04 Å. The dispersion of Mo species on CNT/NF was also investigated using HAADF-STEM and XAFS. HAADF-STEM images reveal numerous isolated bright spots corresponding to individual Mo atoms (Figure S6c), with no detectable Mo clusters or nanoparticles. The Mo K-edge XANES spectra indicate that the absorption edge lies between that of Mo foil and MoO₃,

confirming that Mo is in an oxidized state (Figure S6a). Moreover, the Fourier transform (FT) k²-weighted EXAFS spectra exhibit a primary peak at ~1.38 Å (Figure S6b), assigned to Mo–C coordination, and show no Mo–Mo metallic bonding peaks in R-space [44], confirming the absence of Mo–Mo coordination and demonstrating that Mo atoms are atomically dispersed on the CNT/NF support. The absence of crystalline phases corresponding to Mo or Fe in the X-ray diffraction (XRD) pattern further confirms the atomic dispersion of isolated Fe and Mo species within the Fe SACs/CNT/NF and Mo SACs/CNT/NF catalysts (Figure 3d).

The electronic interactions of surface elements in the catalysts were also systematically investigated through X-ray photoelectron spectroscopy (XPS). As shown in Figure 3e,f, the XPS spectra demonstrate that Fe and Mo predominantly exist in oxidized states rather than in the metallic state. For Fe SACs/CNT/NF, the peaks of Fe 2p_{3/2} and Fe 2p_{1/2} at 710.5 and 723.6 eV are assigned to Fe²⁺, while the peaks at 714.1 and 727.2 eV are attributed to Fe³⁺ species [45, 46]. For Mo SACs/CNT/NF, the Mo 3d spectrum can be deconvoluted into three sets of doublets located at 228.8/231.9, 230.5/233.7, and 232.6/235.7 eV, corresponding to Mo⁴⁺, Mo⁵⁺, and Mo⁶⁺, respectively [14, 47, 48]. The negative shift of the C 1s binding energies after implantation collectively suggests a reconstruction of the interfacial electronic structure, indicating electron transfer from the metal to the carbon framework. Such interfacial coupling, arising from the mismatch in Fermi levels between the metal and carbon substrate, results in significant charge redistribution at the active sites [49], which in turn

reinforces the Fe—C/Mo—C bonding (Figure S7). The ion implantation process increases the degree of sp^3 hybridization and generates a high density of unsaturated defect sites [50], which can act as efficient ‘traps’ for metal atoms, thereby forming exceptionally strong metal-carbon bonds that significantly enhance the stability and dispersion of catalytic sites [51–53]. These results demonstrate that interfacial electronic reconstruction and strong metal-carbon coupling induced by ion implantation underpin the high stability and dispersion of the single-atom catalysts.

2.2 | Oxygen Evolution Reaction Performance and Catalytic Mechanisms

The catalytic activity of Fe SACs/CNT/NF toward OER was evaluated in 1.0 M KOH solution. Linear sweep voltammetry (LSV) curves for both OER and HER were recorded with 85% iR correction, while the uncorrected curves are provided in Figure S8. As shown in Figure 4a, Fe SACs/CNT/NF exhibits superior alkaline OER performance compared to CNT/NF and RuO_2/NF , achieving an overpotential of only 260 mV at a current density of 10 mA cm^{-2} and an overpotential of 320 mV at a current density of 100 mA cm^{-2} , outperforming CNT/NF (405, 654 mV) and RuO_2/NF (385, 550 mV). The Tafel plot derived from the polarization curves was further used to assess the catalytic kinetics of OER. In Figure 4b, Fe SACs/CNT/NF demonstrates a Tafel slope of $46.75 \text{ mV dec}^{-1}$, significantly lower than that of CNT/NF ($112.11 \text{ mV dec}^{-1}$) and RuO_2/NF ($84.19 \text{ mV dec}^{-1}$), indicating faster OER kinetics. Electrochemical impedance spectroscopy (EIS) further reveals that the charge transfer resistance (R_{ct}) of Fe SACs/CNT/NF is lower than that of commercial RuO_2/NF (Figure 4c), suggesting rapid electron transfer during the water oxidation process. Moreover, Fe SACs/CNT/NF demonstrate the lowest overpotential and Tafel slope compared with other commonly reported metal-based single-atom catalysts (Table S4), highlighting its superior intrinsic catalytic activity and efficient reaction kinetics.

The electrochemical active surface area (ECSA) was estimated from the double-layer capacitance (C_{dl}). Cyclic voltammetry (CV) testing was conducted at different scan rates ($20\text{--}120 \text{ mV s}^{-1}$) in the non-Faradaic region. The calculated C_{dl} for Fe SACs/CNT/NF and RuO_2/NF were 43.2 and 27.1 mF cm^{-2} , respectively, indicating that Fe SACs/CNT/NF possesses a higher density of exposed active sites compared to RuO_2/NF (Figure S9). A larger ECSA implies more active sites available for electrocatalytic reactions, thereby enhancing OER activity. Additionally, intrinsic activity was evaluated using turnover frequencies (TOF). At an overpotential of 350 mV during the OER, the TOF value for Fe SACs/CNT/NF is 7.86 times that of RuO_2/NF , revealing the remarkably enhanced intrinsic activity of Fe single-atom sites and their efficient utilization under alkaline conditions (Figure S10). Besides, the durability of catalysts is a critical metric for evaluating their performance. Chronopotentiometry was employed to monitor the OER current density at the overpotential of 400 mV (Figure 4d). The result demonstrates the robust durability of Fe SACs/CNT/NF, capable of continuous operation for over 100 h at 100 mA cm^{-2} . Additionally, accelerated stability tests (inset of Figure 4d) demonstrated excellent electrochemical durability, with a negligible overpotential increase of only 5 mV

at 10 mA cm^{-2} after 6000 CV cycles. The experimental yield of O_2 was measured by the gas collection method at 200 mA cm^{-2} , showing 95.1% Faradaic efficiency for Fe SACs/CNT/NF in Figure S11. The XPS spectra (Figure S12) reveal that the fraction of Fe^{3+} increases after OER cycling, suggesting that the surface Fe species are further oxidized during the reaction. The SEM images (Figure S13a,b), aberration-corrected HAADF-STEM image (Figure S13c), and corresponding STEM and EDS mapping images (Figure S13d) display that the morphology and structure of the Fe SACs/CNT/NF remain well-preserved after the cycling test, confirming its robustness toward water electrolysis. ICP-OES analysis exhibits the exceptional structural stability of Fe SACs/CNT/NF catalysts, with negligible metal leaching detected after prolonged testing (Table S5). The metal leaching process is influenced by the catalyst surface structure, electrolyte chemistry, and operational conditions. In alkaline media, loosely bound Fe and Mo atoms at defect sites may undergo surface oxidation or hydroxylation, leading to minor dissolution, while the majority of single-atom sites are strongly anchored to the carbon support via stable Fe—C or Mo—C bonds [54]. Following this initial minor leaching, the remaining metal atoms achieve a thermodynamically and electrochemically stable state [55], accounting for the retention of active sites and the high long-term catalytic stability observed experimentally (Figure 4d). It should be noted that any Ni present originates solely from the NF substrate. ICP-OES analysis after long-term OER testing detected only 0.02 wt.% Ni relative to the $\sim 60 \text{ mg}$ NF, indicating negligible Ni leaching. As shown in Figures S14a,b, while partial surface oxidation of NF can generate NiOOH -like species, control experiments with bare NF show a much higher overpotential (475 mV) at 10 mA cm^{-2} compared with Fe SACs/CNT/NF (260 mV), demonstrating that the substantially enhanced OER activity primarily arises from the introduced Fe single-atom sites rather than Ni leaching or NiOOH formation.

Current research generally indicates that the catalytic efficiency of the OER strongly depends on the adsorption behavior of oxygen intermediates on the catalyst surface [56]. The electronic band structure of the catalyst surface, particularly its alignment with the p-orbitals of oxygen intermediates, is crucial for enhancing catalytic performance [57]. To this end, this study employs density functional theory (DFT) calculations to systematically analyze the adsorption configurations of key intermediates and the changes in their Gibbs free energy during the OER process. It reveals that an alternative mechanism [58] with 2 OH groups initially adsorbed on the Fe site takes place (red pathway in Figure 4f), instead of the traditional $\ast\text{O}$ pathway (blue pathway in Figure 4f). The conventional OER pathway undergoes a high-energy barrier of 3.08 eV for the $\ast\text{O}$ to $\ast\text{OOH}$ conversion (Figure S15), whereas the proposed 2OH OER pathway proceeds through a proton-retained $\ast\text{OH}\text{--OH}$ intermediate and significantly lowers the barrier to 1.98 eV (Figure 4e,f), indicating facilitated O—O coupling kinetics. Such hydroxyl-assisted intermediates have also been reported to be energetically more favorable than the conventional pathway on SAC systems [59]. Moreover, in situ Raman spectroscopy reveals the dynamic evolution of the $\ast\text{OOH}$ intermediate during OER (Figure S16) [60, 61], and together with the DFT results, supports the proposed hydroxyl-assisted 2OH OER pathway. Furthermore, the electronic structure of Fe in C-Fe was analyzed. The projected density of states (PDOS) results in Figure 4g show that Fe exhibits a high density of

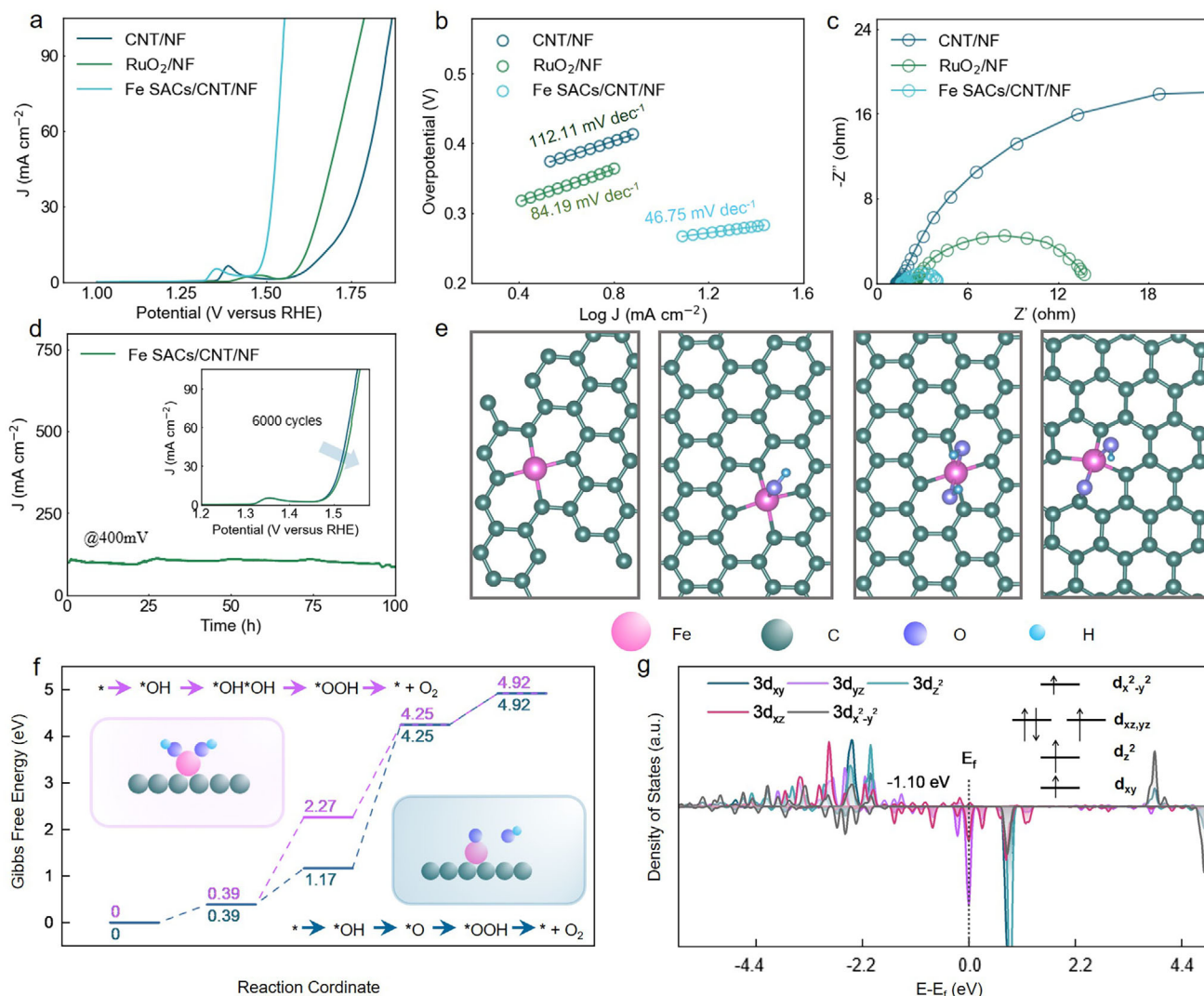


FIGURE 4 | Electrocatalytic performance for OER. (a) OER polarization curves at a scan rate of 5 mV s⁻¹ and (b) Tafel slope (c) EIS measurements at the potential corresponding to a current density of 10 mA cm⁻² of CNT/NF, Fe SACs/CNT/NF, RuO₂/NF. (d) OER chronoamperometry curve of Fe SACs/CNT/NF at an overpotential of 400 mV. The inset curves are LSV curves before and after the stability test. (e) The optimal reaction pathway and adsorption of oxygen intermediates on Fe SACs/CNT/NF. (f) The Gibbs free energies of Fe SACs/CNT/NF through different reaction pathways. (g) Projected density of states (PDOS) plots of d-orbitals of Fe in Fe SACs/CNT/NF.

unoccupied d-orbitals near the Fermi level. These unoccupied states facilitate strong hybridization with the oxygen p-orbitals of reaction intermediates (e.g., *OOH), enhancing interfacial charge transfer and significantly accelerating the redox reaction kinetics [62]. Specifically, the d_{yz} and d_{xz} orbitals display pronounced density of states peaks at the Fermi level (E_f), indicating partial electron occupation in the t_{2g} orbitals. This contributes to improved electronic conductivity, which is critical for efficient charge transport during the OER process.

2.3 | Hydrogen Evolution Reaction Performance

The HER performance of Mo SACs/CNT/NF was evaluated under the same conditions as those used for OER. The Mo SACs/CNT/NF displays outstanding alkaline HER activity, requiring overpotentials of only 14 mV at a current density of 10 mA cm⁻² and 74 mV at 100 mA cm⁻² (Figure 5a). These values

significantly outperform those of commercial 20 wt.% Pt/C/NF (34 mV at 10 mA cm⁻² and 155 mV at 100 mA cm⁻²). The Tafel slope of Mo SACs/CNT/NF is 28.04 mV dec⁻¹ (Figure 5b), significantly lower than that of commercial 20 wt.% Pt/C/NF (48.58 mV dec⁻¹), suggesting that the HER follows the Volmer-Tafel mechanism with the Tafel step being the rate-determining step [63]. EIS further reveals that the R_{ct} of Mo SACs/CNT/NF is lower than that of 20 wt.% Pt/C/NF (Figure 5c), indicating its superior charge transfer capability. In situ Raman spectra of Mo SACs/CNT/NF show that interfacial water molecules exhibit potential-dependent behavior (Figure S17). Raman peak shifts indicate that the signals originate from water layers near the electrode surface, reflecting strong water adsorption on Mo SACs/CNT/NF. Intensity analysis suggests that the interfacial water is predominantly 1H-down, with its signal increasing at more negative potentials, indicating a close involvement in the HER process [64]. The durability of the catalysts was also assessed. As shown in Figure 5d, the current density maintains

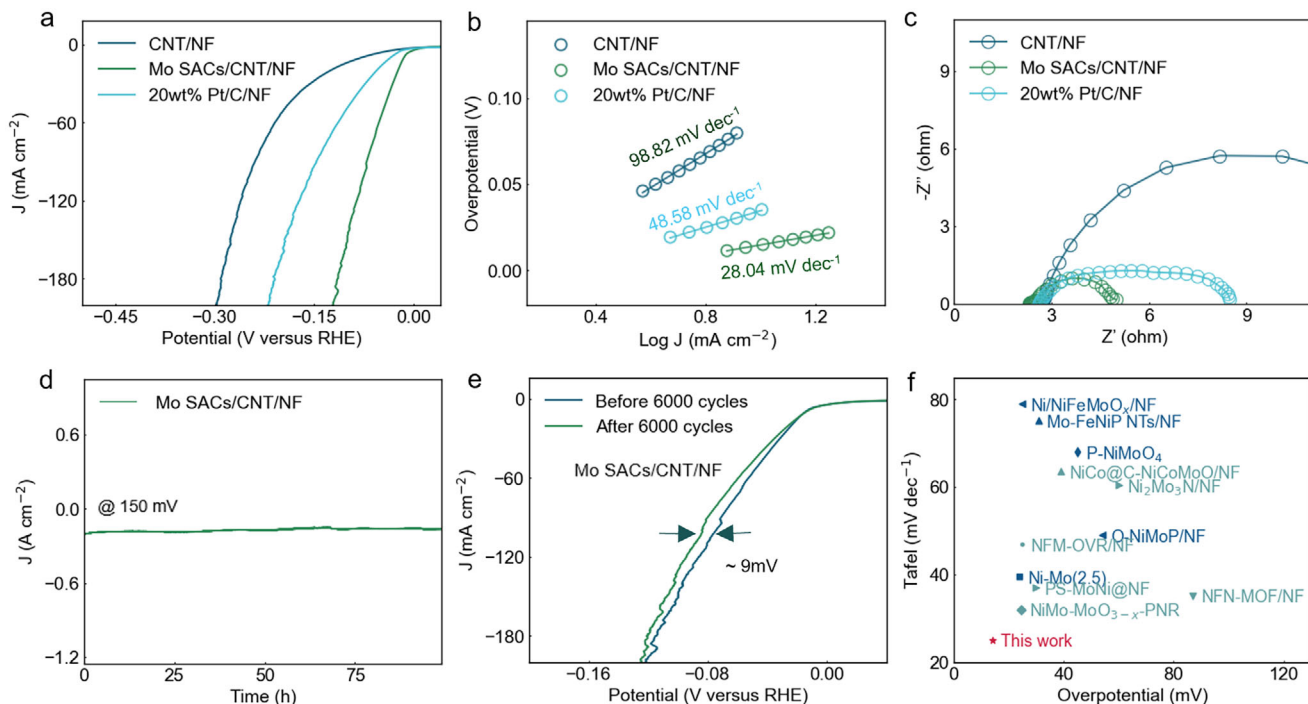


FIGURE 5 | Electrocatalytic performance for HER. (a) HER polarization curves at a scan rate of 5 mV s⁻¹, (b) Tafel slope, and (c) EIS curves at a current of 10 mA cm⁻² of CNT/NF, Mo SACs/CNT/NF, 20 wt.% Pt/C in 1 M KOH. (d) HER chronoamperometry curves of Mo SACs/CNT/NF at 150 mV. (e) The HER LSV curves before and after the stability test. (f) Comparison of the overpotential and Tafel slope of Mo SACs/CNT/NF and recently reported catalysts at 10 mA cm⁻² in 1.0 M KOH.

stability after 100 h of continuous operation, while accelerated stability tests revealed that the overpotential at 100 mA cm⁻² increased by merely 9 mV after 6000 CV cycles (Figure 5e). Similarly, the Faradaic efficiency for HER was determined at 200 mA cm⁻² via the same gas collection approach, yielding a value of 97.4% for Mo SACs/CNT/NF (Figure S18). Post-cycling negative shifts in Mo binding energies suggest the reduction of high-valence states during HER, as shown in Figure S19a [65]. Meanwhile, the STEM and EDS mapping images of Mo SACs/CNT/NF presented in Figure S19b support the structural stability of the ion-implanted catalysts. ICP-OES analysis also reveals outstanding structural stability of the Mo SACs/CNT/NF catalysts, with negligible metal leaching after extended testing (Table S6). When compared to recently reported catalysts supported on NF, as exhibited in Figure 5f and Table S7, Mo SACs/CNT/NF exhibit an outstanding intrinsic activity for alkaline HER, positioning them among the most advanced non-noble metal alkaline HER catalysts to date.

2.4 | Overall Water Splitting Performance

Benefiting from their outstanding HER and OER activities, the overall water-splitting performance of the system consisting of Mo SACs/CNT/NF (cathode) and Fe SACs/CNT/NF (anode) was further evaluated using an anion exchange membrane (AEM) electrolyzer (Figure 6a,b). LSV measurements were conducted using the Mo SACs/CNT/NF||Fe SACs/CNT/NF pair in 1 M KOH at both room-temperature and 60°C. The Mo SACs/CNT/NF||Fe SACs/CNT/NF shows higher activity than the pair with commercial catalysts at room-temperature. Only 1.66 V was required to

deliver 500 mA cm⁻² at 60°C, which is lower than that of the noble-metal benchmark (20 wt.% Pt/C/NF||RuO₂/NF) (Figure 6c and Figure S20). Figure 6d and Table S8 demonstrate the superior electrocatalytic activity of our catalyst compared to existing counterparts. The excellent stability of Mo SACs/CNT/NF||Fe SACs/CNT/NF was further confirmed by a chronoamperometric stability test (Figure 6e), which shows stable water splitting performance for 100 h at 500 mA cm⁻² with negligible attenuation.

These findings demonstrate that the rationally designed Mo SACs/CNT/NF and Fe SACs/CNT/NF catalysts exhibit remarkable intrinsic activity and durability for HER and OER, respectively. Their synergistic integration in a two-electrode configuration achieves superior overall water-splitting performance, even surpassing that of benchmark noble-metal-based catalysts. This work highlights the strategy of anchoring atomically dispersed transition-metal sites on conductive CNT/NF scaffolds as a promising approach for developing next-generation non-noble-metal electrocatalysts.

3 | Conclusion

In conclusion, the Fe SACs/CNT/NF and Mo SACs/CNT/NF catalysts, constructed via ion implantation, demonstrate exceptional performance for overall water splitting. These catalysts feature uniformly anchored and highly accessible single-atom metal sites on a conductive CNT/NF scaffold. The optimized tailored M–C coordination environment endows the active sites with favorable electronic structures, which enhance intermediate adsorption/desorption kinetics and facilitate efficient electron

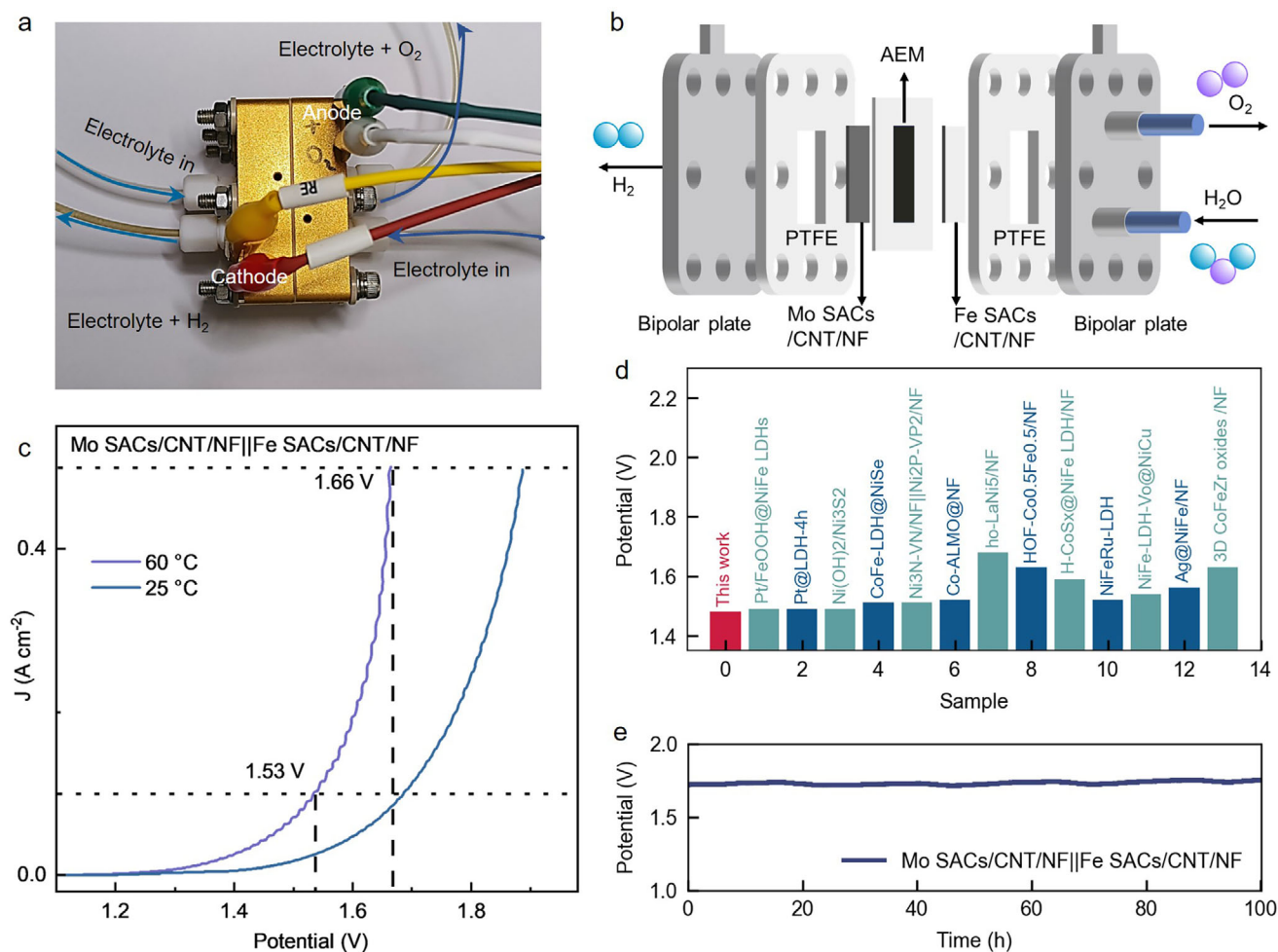


FIGURE 6 | Electrocatalytic performance for overall water-splitting. (a) Flow-type anion exchange membrane (AEM) electrolyzer device and (b) the specific structure of AEM. (c) LSV curves with a 4 cm² electrode area of as-prepared electrodes in 1.0 M KOH at room-temperature and 60 °C, respectively. (d) Overall water splitting potentials of Mo SACs/CNT/NF||Fe SACs/CNT/NF compared with recently reported catalysts at 10 mA cm⁻² in 1.0 M KOH. (e) Chronopotentiometry curve in 1.0 M KOH at 60 °C of Mo SACs/CNT/NF || Fe SACs/CNT/NF at a current density of 500 mA cm⁻².

transfer. As a result, the AEM electrolyzer employing these catalysts achieves a low cell voltage of 1.66 V to deliver 500 mA cm⁻² at 60 °C, surpassing the performance of the noble-metal benchmark (20 wt.% Pt/C/NF||RuO₂/NF) and exhibiting remarkable durability for over 100 h of continuous operation. Furthermore, the developed catalysts show significant economic advantages, with material costs reduced by approximately 41% compared to precious metal systems. This work establishes an effective class of high-performance, durable, and non-precious metal single-atom catalysts, providing a practical path paving the way for practical and efficient energy conversion.

4 | Experimental Section

4.1 | Chemicals

Multi-walled carbon nanotubes (MWCNTs) with a purity exceeding 95%, lengths ranging from 0.5 to 2 μm, and diameters less than 8 nm were procured from XFNANO. The 20 wt.% platinum on carbon (Pt/C) catalyst was sourced from MACKLIN. Additional chemicals, including dicalcium phosphate (KCl, solid, > 99.5%),

potassium hydroxide (KOH, solid, > 85%), and isopropyl alcohol ((CH₃)₂CHOH, solution, > 99.7%), were obtained from TGREAG. Nickel foam substrates were supplied by MTI Corporation. The Nafion D-520 dispersion (5 wt.%) was purchased from GZAMHOO.

4.2 | Catalyst Synthesis via Ion Implantation

Commercial nickel foam (NF) substrates were cut into pieces measuring 1 × 1.5 cm². To remove surface oxides and organic contaminants, the NF pieces were ultrasonically cleaned sequentially in absolute ethanol, 1 M hydrochloric acid (HCl) solution, and deionized (DI) water for 30 min per solvent. During the DI water cleaning step, the water was replaced every 3 min to ensure thorough removal of any residual HCl. Subsequently, a final 1 min ultrasonic rinse in ethanol was performed to facilitate rapid drying. The cleaned NF substrates were then dried under vacuum at 80 °C for 10 h. The catalyst ink was prepared by uniformly dispersing 6 mg of multi-walled carbon nanotubes (MWCNTs) in a mixed solvent containing 1100 μL of isopropyl alcohol (IPA) and 100 μL of Nafion D-520 dispersion (5 wt.%) via ultrasonication for

30 min. A 200 μL aliquot of this homogeneous ink was then drop-cast onto the pretreated NF substrate ($1 \times 1 \text{ cm}^2$) and dried at room temperature to form the precursor electrodes. The metal active sites (Fe, Mo) were subsequently implanted into the CNT/NF precursor using a customized ion implantation system. For each metal species, the implantation was performed at a fixed energy of 10 keV and with a dose of 1×10^{17} ions cm^{-2} .

The details of materials characterization, electrochemical measurements, DFT calculations, and techno-economic analysis are presented in the [Supporting Information](#).

Author Contributions

Yang Xu: Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft & review, Visualization. Yue Li: Writing – review & editing, Supervision, Validation. Yunbiao Zhao: Formal analysis, Data curation. Xiner Chen: Formal analysis, Data curation. Jin Zhang: Writing – review & editing, Supervision, Funding acquisition. Xueting Feng: Writing – review & editing, Supervision, Funding acquisition. Ziqiang Zhao: Writing – review & editing, Supervision, Project administration, Funding acquisition.

Acknowledgements

This work was financially supported by the National Natural Science Foundation of China (Grant Nos. 52272033, 22494641, 5201006 and 52402041), the Beijing National Laboratory for Molecular Sciences (BNLMS-CXTD-202001) and the Shenzhen Science and Technology Innovation Commission (KQTD20221101115627004).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. I. Staffell, D. Scamman, A. Velazquez Abad, et al., “The Role of Hydrogen and Fuel Cells in the Global Energy System,” *Energy & Environmental Science* 12, no. 2 (2019): 463–491, <https://doi.org/10.1039/C8EE01157E>.
2. N. Esfandiari, M. Aliofkhaezrai, A. N. Colli, et al., “Metal-Based Cathodes for Hydrogen Production by Alkaline Water Electrolysis: Review of Materials, Degradation Mechanism, and Durability Tests,” *Progress in Materials Science* 144 (2024): 101254, <https://doi.org/10.1016/j.pmatsci.2024.101254>.
3. Y. Yang, P. Li, X. Zheng, et al., “Anion-Exchange Membrane Water Electrolyzers and Fuel Cells,” *Chemical Society Reviews* 51, no. 23 (2022): 9620–9693, <https://doi.org/10.1039/D2CS00038E>.
4. X. Xu, K. Guo, J. Sun, et al., “Interface Engineering of Mo-Doped $\text{Ni}_2\text{P}/\text{Fe}_x\text{P-V}$ Multiheterostructure for Efficient Dual-pH Hydrogen Evolution and Overall Water Splitting,” *Advanced Functional Materials* 34, no. 33 (2024): 2400397, <https://doi.org/10.1002/adfm.202400397>.
5. Y. Li, Y. Sun, Y. Qin, et al., “Recent Advances on Water-Splitting Electrocatalysis Mediated by Noble-Metal-Based Nanostructured Materials,” *Advanced Energy Materials* 10, no. 11 (2020): 1903120, <https://doi.org/10.1002/aenm.201903120>.
6. J. Yin, J. Jin, M. Lu, et al., “Iridium Single Atoms Coupling With Oxygen Vacancies Boosts Oxygen Evolution Reaction in Acid Media,” *Journal of*

the American Chemical Society 142, no. 43 (2020): 18378–18386, <https://doi.org/10.1021/jacs.0c05050>.

7. M. Tahir, L. Pan, F. Idrees, et al., “Electrocatalytic Oxygen Evolution Reaction for Energy Conversion and Storage: A Comprehensive Review,” *Nano Energy* 37 (2017): 136–157, <https://doi.org/10.1016/j.nanoen.2017.05.022>.
8. L. Han, S. Dong, and E. Wang, “Transition-Metal (Co, Ni, and Fe)-Based Electrocatalysts for the Water Oxidation Reaction,” *Advanced Materials* 28, no. 42 (2016): 9266–9291, <https://doi.org/10.1002/adma.201602270>.
9. R. D. L. Smith, M. S. Prévot, R. D. Fagan, et al., “Photochemical Route for Accessing Amorphous Metal Oxide Materials for Water Oxidation Catalysis,” *Science* 340, no. 6128 (2013): 60–63, <https://doi.org/10.1126/science.1233638>.
10. Y. Wang, L. Ouyang, S. Zhang, et al., “Sulfur Vacancy-Rich ZnIn_2S_4 via Solvent Engineering: Unveiling Structure-Defect-Activity Relationships for Photocatalytic Ranitidine Degradation,” *Chemical Engineering Journal* 526 (2025): 170917, <https://doi.org/10.1016/j.cej.2025.170917>.
11. Z. Xie, F. Li, C. Belver, et al., “Surface and Morphology Modulated Adsorption-Activation of Trace Concentration Hydrogen Peroxide With Multiple Electron Transfer Pathways and Sustained Fenton Reactions,” *Applied Catalysis B: Environment and Energy* 364 (2025): 124840, <https://doi.org/10.1016/j.apcatb.2024.124840>.
12. M. Li, H. Zhu, Q. Yuan, et al., “Proximity Electronic Effect of Ni/Co Diatomic Sites for Synergistic Promotion of Electrocatalytic Oxygen Reduction and Hydrogen Evolution,” *Advanced Functional Materials* 33, no. 4 (2023): 2210867, <https://doi.org/10.1002/adfm.202210867>.
13. J. Sui, H. Liu, S. Hu, et al., “A General Strategy to Immobilize Single-Atom Catalysts in Metal–Organic Frameworks for Enhanced Photocatalysis,” *Advanced Materials* 34, no. 6 (2022): 2109203, <https://doi.org/10.1002/adma.202109203>.
14. X. Sun, B. Ge, S. Xue, et al., “Lattice-Matched $\text{Cu}_3\text{Pd}@-\alpha\text{-MoC}_{1-x}$ Heterointerfaces Boost Alkaline Hydrogen Evolution by Accelerating Water Dissociation and Optimizing Hydrogen Adsorption Energy,” *Applied Catalysis B: Environment and Energy* 386 (2026): 126418, <https://doi.org/10.1016/j.apcatb.2026.126418>.
15. X. Sun, J. Liu, Y. Du, et al., “Bifunctional Arrays of NiRu Single-Atom Alloy Nanoparticles Confined in a Porous Carbon Nanosheet for Sustained Anion-Exchange Membrane Water Electrolysis,” *Nano Letters* 25, no. 30 (2025): 11680–11688, <https://doi.org/10.1021/acs.nanolett.5c02754>.
16. L. Ran, Y. Xu, Y. Zhang, et al., “Engineering Iridium–Ruthenium Dual-Atomic Active Sites on Redox-Active Covalent Organic Frameworks for Boosted Overall Water Splitting,” *Small* 22, no. 13 (2026): 10427, <https://doi.org/10.1002/smll.202510427>.
17. W. Li, C. Liu, C. Gu, J. H. Choi, S. Wang, and J. Jiang, “Interlayer Charge Transfer Regulates Single-Atom Catalytic Activity on Electride/Graphene 2D Heterojunctions,” *Journal of the American Chemical Society* 145, no. 8 (2023): 4774–4783, <https://doi.org/10.1021/jacs.2c13596>.
18. X. Lin, W. Hu, J. Xu, et al., “Alleviating OH Blockage on the Catalyst Surface by the Puncture Effect of Single-Atom Sites to Boost Alkaline Water Electrolysis,” *Journal of the American Chemical Society* 146, no. 7 (2024): 4883–4891, <https://doi.org/10.1021/jacs.3c13676>.
19. R. Wang, J. Guo, J. Li, et al., “Transforming Single-Atom Site to Dual-Atom Site in Fe–N–C Catalysts: A Universal Strategy for Enhancing Durability in Proton-Exchange Membrane Fuel Cells,” *Angewandte Chemie International Edition* 64, no. 34 (2025): 202510671, <https://doi.org/10.1002/anie.202510671>.
20. W. Chen, G. Qian, H. Wang, et al., “Rationalizing the d-Band Model From Theory to Practice in Catalyst Design,” *Journal of the American Chemical Society* 147, no. 51 (2025): 46729–46744, <https://doi.org/10.1021/jacs.5c17673>.
21. L. Qi and J. Guan, “Orbital Hybridizations in Single-Atom Catalysts for Electrocatalysis,” *Science Bulletin* 70, no. 11 (2025): 1856–1871, <https://doi.org/10.1016/j.scib.2025.04.009>.

22. X. W. Lv, J. Gong, S. Wang, et al., "Engineering Orbital Hybridization in Advanced Electrocatalysts for Energy Conversion: Fundamentals, Modulations, and Perspectives," *Advanced Energy Material* (2025), 2501129, <https://doi.org/10.1002/aenm.202501129>.
23. Y. Zhang, F. Chen, X. Yang, et al., "Electronic Metal-Support Interaction Modulates Cu Electronic Structures for CO₂ Electroreduction to Desired Products," *Nature Communications* 16, no. 1 (2025): 1956, <https://doi.org/10.1038/s41467-025-57307-6>.
24. Y. Chen, S. Jiang, X. Liu, et al., "Deciphering Electronic Metal-Support Interaction in CeI/ZnO Single-Atom Catalysts for Boosting CO₂ Electroreduction," *Applied Catalysis B: Environment and Energy* 384 (2026): 126178, <https://doi.org/10.1016/j.apcatb.2025.126178>.
25. C. Ye, J. Liu, Q. Zhang, et al., "Activating Metal Oxides Nanocatalysts for Electrocatalytic Water Oxidation by Quenching-Induced Near-Surface Metal Atom Functionality," *Journal of the American Chemical Society* 143, no. 35 (2021): 14169–14177, <https://doi.org/10.1021/jacs.1c04737>.
26. Y. Zhu, X. He, and Y. Mo, "First Principles Study on Electrochemical and Chemical Stability of Solid Electrolyte–Electrode Interfaces in All-Solid-State Li-Ion batteries," *Journal of Materials Chemistry A* 4, no. 9 (2016): 3253–3266, <https://doi.org/10.1039/C5TA08574H>.
27. J. Wang, H. Kong, J. Zhang, Y. Hao, Z. Shao, and F. Ciucci, "Carbon-Based Electrocatalysts for Sustainable Energy Applications," *Progress in Materials Science* 116 (2021): 100717, <https://doi.org/10.1016/j.pmatsci.2020.100717>.
28. Y. Li, M. Lu, Y. Wu, H. Xu, and J. Gao, "Trimetallic Metal–Organic Framework Derived Carbon-Based Nanoflower Electrocatalysts for Efficient Overall Water Splitting," *Advanced Materials Interfaces* 6, no. 12 (2019): 1900290, <https://doi.org/10.1002/admi.201900290>.
29. R. Paul, Q. Zhai, A. K. Roy, and L. Dai, "Charge Transfer of Carbon Nanomaterials for Efficient Metal-Free Electrocatalysis," *Interdisciplinary Materials* 1, no. 1 (2022): 28–50, <https://doi.org/10.1002/idm2.12010>.
30. Y. Li, Y. Liu, Y. Zhao, et al., "Highly-Defective Graphene as a Metal-Free Catalyst for Chemical Vapor Deposition Growth of Graphene Glass," *Carbon* 187 (2022): 272–279, <https://doi.org/10.1016/j.carbon.2021.11.019>.
31. H. J. Qiu, Y. Ito, W. Cong, et al., "Nanoporous Graphene With Single-Atom Nickel Dopants: An Efficient and Stable Catalyst for Electrochemical Hydrogen Production," *Angewandte Chemie International Edition* 54, no. 47 (2015): 14031–14035, <https://doi.org/10.1002/anie.201507381>.
32. M. Ning, Y. Wang, L. Wu, et al., "Hierarchical Interconnected NiMoN With Large Specific Surface Area and High Mechanical Strength for Efficient and Stable Alkaline Water/Seawater Hydrogen Evolution," *Nano-Micro Letters* 15, no. 1 (2023): 157, <https://doi.org/10.1007/s40820-023-01129-y>.
33. Q. Zhang, P. Li, D. Zhou, Z. Chang, Y. Kuang, and X. Sun, "Superaerophobic Ultrathin Ni–Mo Alloy Nanosheet Array From In Situ Topotactic Reduction for Hydrogen Evolution Reaction," *Small* 13, no. 41 (2017): 1701648, <https://doi.org/10.1002/sml.201701648>.
34. J. Zhang, T. Wang, P. Liu, et al., "Efficient Hydrogen Production on MoNi₄ Electrocatalysts With Fast Water Dissociation Kinetics," *Nature Communications* 8, no. 1 (2017): 15437, <https://doi.org/10.1038/ncomms15437>.
35. Y. Li, H. Zhang, M. Jiang, Q. Zhang, P. He, and X. Sun, "3D Self-Supported Fe-Doped Ni₂P Nanosheet Arrays as Bifunctional Catalysts for Overall Water Splitting," *Advanced Functional Materials* 27, no. 37 (2017): 1702513, <https://doi.org/10.1002/adfm.201702513>.
36. X. Zou, Y. Liu, G. D. Li, et al., "Ultrafast Formation of Amorphous Bimetallic Hydroxide Films on 3D Conductive Sulfide Nanoarrays for Large-Current-Density Oxygen Evolution Electrocatalysis," *Advanced Materials* 29, no. 22 (2017): 1700404, <https://doi.org/10.1002/adma.201700404>.
37. J. Kim, S. H. Her, D. S. Kim, et al., "Stabilized Co Single-Atom Catalyst via Ion Implantation for Efficient Hydrogen Production," *Small* 21, no. 34 (2025): 2505383, <https://doi.org/10.1002/sml.202505383>.
38. Y. Xie, Y. Li, Z. Peng, et al., "Nano-Seeding Catalysts for High-Density Arrays of Horizontally Aligned Carbon Nanotubes With Wafer-Scale Uniformity," *Nature Communications* 16, no. 1 (2025): 149, <https://doi.org/10.1038/s41467-024-55515-0>.
39. Y. Li, Y. Xu, Y. Zhao, et al., "A Non-Equilibrium Strategy for the General Synthesis of Single-Atom Catalysts," *arXiv* (2026), arXiv:2601.12477, <https://doi.org/10.48550/arXiv.2601.12477>.
40. X. Ji, X. Yang, Y. Liu, et al., "Wafer-Scale Manufacturing of Ultra-Broadband, High-Power Erbium-Doped Integrated Lasers," *Nature Communications* 17, no. 1 (2026): 3722, <https://doi.org/10.1038/s41467-026-69787-1>.
41. P. Chen, T. Zhou, L. Xing, et al., "Atomically Dispersed Iron–Nitrogen Species as Electrocatalysts for Bifunctional Oxygen Evolution and Reduction Reactions," *Angewandte Chemie International Edition* 56, no. 2 (2017): 610–614, <https://doi.org/10.1002/anie.201610119>.
42. L. Jiao, G. Wan, R. Zhang, H. Zhou, S. H. Yu, and H. L. Jiang, "From Metal–Organic Frameworks to Single-Atom Fe Implanted N-doped Porous Carbons: Efficient Oxygen Reduction in Both Alkaline and Acidic Media," *Angewandte Chemie International Edition* 57, no. 28 (2018): 8525–8529, <https://doi.org/10.1002/anie.201803262>.
43. M. Zhang, Y. G. Wang, W. Chen, et al., "Metal (Hydr)oxides@Polymer Core–Shell Strategy to Metal Single-Atom Materials," *Journal of the American Chemical Society* 139, no. 32 (2017): 10976–10979, <https://doi.org/10.1021/jacs.7b05372>.
44. G. Meng, W. Lan, L. Zhang, et al., "Synergy of Single Atoms and Lewis Acid Sites for Efficient and Selective Lignin Disassembly Into Monolignol Derivatives," *Journal of the American Chemical Society* 145, no. 23 (2023): 12884–12893, <https://doi.org/10.1021/jacs.3c04028>.
45. X. F. Lu, L. Yu, and X. W. Lou, "Highly Crystalline Ni-Doped FeP/Carbon Hollow Nanorods as all-pH Efficient and Durable Hydrogen Evolving Electrocatalysts," *Science Advances* 5, no. 2 (2019): aav6009, <https://doi.org/10.1126/sciadv.aav6009>.
46. Y. Wu, M. Chen, H. Sun, et al., "Coupling Ir Single Atom With NiFe LDH/NiMo Heterointerface Toward Efficient and Durable Water Splitting at Large Current Density," *Applied Catalysis B: Environment and Energy* 360 (2025): 124548, <https://doi.org/10.1016/j.apcatb.2024.124548>.
47. D. Song, J. Roh, J. Choi, et al., "Heterogeneous Structure of Ni–Mo Nanoalloys Decorated on MoO_x for an Efficient Hydrogen Evolution Reaction Using Hydrogen Spillover," *Advanced Science* 11, no. 39 (2024): 2403752, <https://doi.org/10.1002/advs.202403752>.
48. Y. Wang, A. Sharma, T. Duong, et al., "Direct Solar Hydrogen Generation at 20% Efficiency Using Low-Cost Materials," *Advanced Energy Materials* 11, no. 34 (2021): 2101053, <https://doi.org/10.1002/aenm.202101053>.
49. M. Liu, Y. Li, L. Yang, et al., "Defect-Triggered Orbital Hybridization in FeMn Dual-Atom Catalysts Toward Sabatier-Optimized Oxygen Reduction," *Angewandte Chemie International Edition* 64, no. 28 (2025): 202505268, <https://doi.org/10.1002/anie.202505268>.
50. Y. Li, X. Ren, X. Feng, et al., "Atomic Interface Engineering of Battery Current Collectors via Ion Implantation," *arXiv* (2025), arXiv:2508.00236, <https://ui.adsabs.harvard.edu/abs/2025arXiv250800236L>.
51. M. A. Caro, V. L. Deringer, J. Koskinen, T. Laurila, and G. Csányi, "Growth Mechanism and Origin of High sp³ Content in Tetrahedral Amorphous Carbon," *Physical Review Letters* 120, no. 16 (2018): 166101, <https://doi.org/10.1103/PhysRevLett.120.166101>.
52. H. H. Wong, M. Sun, T. Wu, et al., "Neighboring Effect in Single-Atom Catalysts for the Electrochemical Carbon Dioxide Reduction Reaction," *eScience* 4, no. 1 (2024): 100140, <https://doi.org/10.1016/j.esci.2023.100140>.
53. W. Song, C. Xiao, J. Ding, et al., "Review of Carbon Support Coordination Environments for Single Metal Atom Electrocatalysts (SACS)," *Advanced Materials* 36, no. 1 (2024): 2301477, <https://doi.org/10.1002/adma.202301477>.

54. J. Cho, T. Lim, H. Kim, et al., "Importance of Broken Geometric Symmetry of Single-Atom Pt Sites for Efficient Electrocatalysis," *Nature Communications* 14, no. 1 (2023): 3233, <https://doi.org/10.1038/s41467-023-38964-x>.
55. C. Li, X. Yang, Y. Kou, et al., "Low-Level Fe Doping in CoMoO₄ Enhances Surface Reconstruction and Electronic Modulation Creating an Outstanding OER Electrocatalyst for Water Splitting," *Inorganic Chemistry* 64, no. 5 (2025): 2508–2517, <https://doi.org/10.1021/acs.inorgchem.4c05100>.
56. F. Cheng and J. Chen, "Metal–Air Batteries: From Oxygen Reduction Electrochemistry to Cathode Catalysts," *Chemical Society Reviews* 41, no. 6 (2012): 2172–2192, <https://doi.org/10.1039/C1CS15228A>.
57. J. K. Nørskov, J. Rossmeisl, A. Logadottir, et al., "Origin of the Overpotential for Oxygen Reduction at a Fuel-Cell Cathode," *The Journal of Physical Chemistry B* 108, no. 46 (2004): 17886–17892, <https://doi.org/10.1021/jp047349j>.
58. L. Zhong and S. Li, "Unconventional Oxygen Reduction Reaction Mechanism and Scaling Relation on Single-Atom Catalysts," *ACS Catalysis* 10, no. 7 (2020): 4313–4318, <https://doi.org/10.1021/acscatal.0c00815>.
59. I. Barlocco, L. A. Cipriano, G. Di Liberto, and G. Pacchioni, "Does the Oxygen Evolution Reaction Follow the Classical OH*, O*, OOH* Path on Single Atom catalysts?," *Journal of Catalysis* 417 (2023): 351–359, <https://doi.org/10.1016/j.jcat.2022.12.014>.
60. G. Li, S. Luan, J. Wu, et al., "Zwitterionic-Molecule-Driven Synergistic Coupling of Dual OER Pathways and Interfacial Water Dynamics for Efficient Alkaline Water Electrolysis," *Advanced Energy Materials* 16, no. 16 (2026): 70749, <https://doi.org/10.1002/aenm.70749>.
61. X. Tao, L. Hou, X. Wang, J. Jin, H. Li, and F. Gao, "Iron and Oxygen Vacancies Co-Modulated Adsorption Evolution and Lattice Oxygen Dual-Path Mechanism for Water Oxidation," *Nature Communications* 16, no. 1 (2025): 8788, <https://doi.org/10.1038/s41467-025-63844-x>.
62. A. Kulkarni, S. Siahrostami, A. Patel, and J. K. Nørskov, "Understanding Catalytic Activity Trends in the Oxygen Reduction Reaction," *Chemical Reviews* 118, no. 5 (2018): 2302–2312, <https://doi.org/10.1021/acs.chemrev.7b00488>.
63. J. Gan, Z. Huang, W. Luo, et al., "Platelet Carbon Nanofibers as Support of Pt-CoO Electrocatalyst for Superior Hydrogen Evolution," *Journal of Energy Chemistry* 52 (2021): 33–40, <https://doi.org/10.1016/j.jechem.2020.04.036>.
64. Q. He, Y. Zhou, L. Mou, et al., "Iron-Doped Ruthenium With a Good Interfacial Environment Achieving Superior Hydrogen Evolution Activity Under Alkaline Conditions," *Energy & Environmental Science* 18, no. 4 (2025): 1984–1991, <https://doi.org/10.1039/D4EE05356G>.
65. J. Li, W. Hu, L. Sun, et al., "Dynamic Self-Reconstruction Heterostructures: Boosting the High-Stability for Hydrogen Evolution of NiMo Alloys in Acidic Environments," *Small* 20, no. 51 (2024): 2403364, <https://doi.org/10.1002/sml.202403364>.

Supporting Information

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